

# Chapter 1

## Spectroscopic Notation

### 1.1 Spectroscopic Notation

This section is intended as a quick review of some of the basics of spectroscopic notation just to get us to the point that we can use Russell-Saunders (RS) term symbols and have some idea of what they mean. The details will be covered later in the course.

First, as a quick reminder, the single-electron eigenstates in atoms are characterized by three quantum numbers. The principal (or radial) quantum number is denoted by  $n$  where  $n$  takes positive integer values. The orbital angular momentum quantum number is denoted by  $l$  where  $l$  takes non-negative integer values (including  $l = 0$ ). The total angular momentum, which is the vector sum of  $l$  and the electron spin, is denoted by quantum number  $j$ , which takes non-negative half-integer values.

#### 1.1.1 Electron configurations and multi-electron atoms

For multi-electron atoms, these individual-electron angular momenta can become correlated between electrons, and we will adopt the convention that capital letters denote operators and quantum numbers that may involve multiple electrons.

For example, it may be reasonable in some circumstances to refer to the *total spin* of the multi-electron system<sup>1</sup>. For instance, a two-electron atom

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<sup>1</sup>The coupling of electron spins to one another in an atom typically arises from the interplay of the Pauli exclusion principle and electrostatic repulsion between the electrons,

may have a total (electron) spin quantum number of  $S = 0$  or  $S = 1$ . The number of orientations allowed is the *spin multiplicity*,  $2S + 1$ , choices for which are given by the number of unique eigenvalues of  $\hat{S}_z$ . This coupling of individual spins  $\mathbf{s}_i$  into a total spin  $\mathbf{S}$  is part of what is called *Russell-Saunders-* or *LS-coupling*.

With apologies in advance, for orbital angular momentum, we are essentially stuck with adopting some letters that get used in atomic spectroscopy to denote particular integer values of the associated quantum number. In particular,  $l = 0$  is denoted by the letter  $s$ ,  $l = 1$  is denoted by the letter  $p$ ,  $l = 2$  is denoted by the letter  $d$ , and  $l = 3$  is denoted by  $f$ . From there on, the letters follow in alphabetical order (except that  $j$  is skipped, and of course the  $p$  and  $s$  that already got used would also be skipped if  $l$  ever got large enough for us to care). So  $l$  taking values of  $l = 0, 1, 2, 3, 4$ , and  $5$  would correspond to the symbols for the value of  $l$  (we could call these  $\ell$ ) being  $\ell = s, p, d, f, g$ , and  $h$  (respectively).

For multi-electron atoms, electrostatic interactions between electrons can give rise to correlations between their orbital angular momenta such that it may be more reasonable to consider the *total electron orbital angular momentum*,  $\mathbf{L} \equiv \sum_i \mathbf{l}_i$ , which is the other part of the Russell-Saunders coupling scheme. The *total* electron orbital angular momentum quantum number  $L$  has values denoted by the same letters as the individual electron orbital angular momentum designations, but capitalized (and in upright font), as in  $\mathcal{L} = S, P, D, F, G, H$ , etc.

The single-electron letters frequently get used when referring to the *electron configuration* in an atom, where the expression

$$n\ell^\alpha \tag{1.1}$$

refers to  $\alpha$  electrons ( $\alpha$  is an integer, and is often not written explicitly when  $\alpha = 1$ ) occupying the  $n\ell$  orbital with principal quantum number  $n$  and electron orbital angular momentum denoted by the letter for  $\ell$ . For example, the ground state electron configuration of hydrogen is  $1s$ , which means it has one electron (the superscript 1 on the  $s$  has been omitted, as discussed above) in an orbital with  $n = 1$  and  $l = 0$ . Helium has two electrons in that orbital in its ground state, so its configuration is  $1s^2$ . Ground state lithium is next, and begins to fill the  $n = 2$  shell, so its configuration is written  $1s^2 2s$ , and so on.

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which will be explored in detail in the chapters on helium)

Filled shells (*i.e.* when all of the available levels corresponding to a given principal quantum number  $n$  are occupied) are sometimes either omitted or written with the corresponding noble gas atom chemical symbol in brackets. Taking ground-state Li as our example, the electron configuration could also be written as [He]2s or just 2s. We will talk later about how these get filled and what all of this means in more detail, but for the moment, we can just take this as a dictionary for reading the electron configurations in the literature.

### 1.1.2 RS term symbols

For multi-electron atoms in the Russell-Saunders coupling scheme, the overall state of the atom is often summarized by a *term symbol* of the form

$$^{2S+1}\mathcal{L}_J \quad (1.2)$$

where  $S$  is the total electron spin,  $\mathcal{L}$  is a capital letter denoting the total electron orbital angular momentum as described above, and  $J$  is the total electron angular momentum. States with odd spatial parity are also often given a following superscript  $o$ . In this course, I will sometimes refer to the whole set of possible values of  $J$  for a given  $L$  and  $S$  as a term, but this language is also sometimes applied to denote specific values of  $J$ .

For example, the ground state of Li, which we noted was described by the electron configuration  $1s^2 2s$ , has  $S = 1/2$ ,  $L = 0$ , and  $J = 1/2$ . So  $2S + 1 = 2$  and  $\mathcal{L} = S$ , allowing us to write the Russell-Saunders term symbol as

$$^2S_{1/2}. \quad (1.3)$$

An atom in a state described by

$$^3P_2^o \quad (1.4)$$

has  $S = 1$  (from the leading superscript 3),  $L = 1$  (from the  $\mathcal{L} = P$ ),  $J = 2$  (from the following subscript 2), and has odd spatial parity (from the following superscript  $o$ ).

When reading a term symbol out loud, the leading superscript denoting the spin multiplicity,  $2S + 1$ , is said using “multiplet” terminology, such as “doublet” for  $2S + 1 = 2$ , “triplet” for  $2S + 1 = 3$ , and then “quartet” for 4, “quintet” for 5, and so on for higher spin multiplicities. So the term symbols in (1.3) and (1.4) would be said as “doublet S one half” and “triplet P two, odd,” respectively.

